

G1 — a GNN distils fixed MDT more faithfully than an MLP

Table 1: Procrustes fidelity to the fixed MDT embedding and the train-AMI consequence. Per-dataset means over five seeds. Paired effect over the twelve dataset means: +0.0028 [−0.0046, 0.0102], 16/60 seed-level wins. **Dead end.**

Dataset	classes	fidelity GNN	fidelity MLP	Δ fidelity	Δ train AMI
MSRC-v5	7	0.9991	0.9999	−0.0008	+0.0045
Handwritten	10	0.9936	0.9935	+0.0001	+0.0024
Wikipedia	10	0.9732	0.9701	+0.0031	−0.0095
UCI	10	0.9905	0.9920	−0.0015	−0.0022
OutdoorScene	8	0.9837	0.9883	−0.0046	+0.0082
Caltech101-7	7	0.9970	0.9989	−0.0019	+0.0078
100Leaves	100	0.5959	0.5567	+0.0392	+0.0066
Yale	15	0.9997	0.9999	−0.0002	−0.0005
3Sources	6	1.0000	1.0000	0.0000	0.0000
BBCSport	5	1.0000	1.0000	0.0000	0.0000
Cora	7	1.0000	1.0000	0.0000	0.0000
CiteSeer	6	1.0000	1.0000	0.0000	−0.0003

G2 — graph message passing improves MDT out-of-sample extension

Table 2: Absolute out-of-sample AMI, per-dataset means over five seeds. **nys** is training-free Nyström, **mlp** is feature-only parametric, **gnn** adds message passing over the same bipartite graph.

Dataset	test AMI			inductive AMI		
	nys	mlp	gnn	nys	mlp	gnn
MSRC-v5	0.6202	0.6892	0.6789	0.5989	0.6176	0.6726
Handwritten	0.8698	0.8756	0.8823	0.8715	0.8792	0.8835
Wikipedia	0.2923	0.2955	0.3458	0.2768	0.2783	0.2958
UCI	0.7818	0.8242	0.8283	0.8047	0.8275	0.8305
OutdoorScene	0.5232	0.5352	0.5495	0.4999	0.5291	0.5470
Caltech101-7	0.5283	0.5404	0.5459	0.5050	0.5176	0.5367
100Leaves	0.6581	0.8591	0.8849	0.6375	0.8618	0.8758
Yale	0.5398	0.4698	0.4895	0.5144	0.4523	0.5004
3Sources	0.0799	0.3535	0.2093	0.0755	0.1316	0.0776
BBCSport	0.1670	0.2231	0.2026	0.1210	0.1143	0.1303
Cora	0.0266	0.0246	0.0256	0.0000	0.0000	0.0000
CiteSeer	0.0258	0.0576	0.0466	0.0082	0.0075	0.0089

Table 3: The decomposition $(\text{GNN} - \text{Nyström}) = (\text{MLP} - \text{Nyström}) + (\text{GNN} - \text{MLP})$. G2 is a claim about the second term only, in bold. Paired 95% Student intervals across dataset means, seeds averaged within a dataset first. **Dead end:** the graph term never clears the +0.02 minimum useful effect.

Contrast	metric	screen (6)	validation (6)	pooled (12)
GNN − Nyström	test	+0.0359 [0.0153, 0.0564]	+0.0602 [−0.0454, 0.1659]	+0.0480 [0.0034, 0.0927]
GNN − Nyström	inductive	+0.0349 [0.0113, 0.0584]	+0.0394 [−0.0632, 0.1420]	+0.0372 [−0.0058, 0.0801]
MLP − Nyström	test	+0.0241 [−0.0033, 0.0515]	+0.0817 [−0.0545, 0.2180]	+0.0529 [−0.0069, 0.1128]
MLP − Nyström	inductive	+0.0154 [0.0047, 0.0261]	+0.0352 [−0.0697, 0.1401]	+0.0253 [−0.0182, 0.0688]
GNN − MLP	test	+0.0118 [−0.0097, 0.0333]	−0.0215 [−0.0872, 0.0442]	−0.0049 [−0.0352, 0.0254]
GNN − MLP	inductive	+0.0195 [−0.0003, 0.0392]	+0.0042 [−0.0308, 0.0393]	+0.0119 [−0.0053, 0.0290]